



Participant's Section

I. Use Case Short Title

AltCredit – Alternative Credit Scoring - The “New-Age” Credit-Score Framework

II. Problem Statement & Background

To get a loan, banks look at your credit history. This credit history is analyzed based on any past or current loan or credit cards that a person holds.

The catch: is if you are a first-time borrower, freelancer, gig worker or a young professional you probably have never taken a formal loan or own a credit card before.

The Result: Because you have no credit history or loan history the bank assume you are a “high risk” candidate and the bank rejects your loan application.

The Impact: Millions of financially responsible people are locked out of the normal banking system and are pushed towards taking loans at a high interest rate via an unorganized sector and fall prey to loan sharks.

Traditional credit scoring models are fundamentally biased against new credit seekers & even the traditional models are not focusing on latest lifestyle changes. These models rely heavily on historical credit data, such as long-term mortgage repayments, existing credit card histories, and established loan aging.

Since the lack of "credit footprint," young loan seekers are often flagged as high-risk by traditional algorithms, even if they are financially disciplined & not considering lifestyle impacts in traditional algorithms. This creates a systemic gap where creditworthy individuals are rejected, preventing them from accessing essential financial tools or inappropriate loan execution. Since, there is a need for an alternative scoring engine (**Rule-Based “New-Age” Credit-Score Framework**) that leverages non-traditional, high-frequency data to provide a fair, transparent, and predictive assessment of creditworthiness.

III. Objective

Create a unified Alternative Credit Scoring platform (**Rule-Based “New-Age” Credit-Score Framework**) that ingests non-traditional data streams to generate a reliable creditworthiness profile for individuals without a formal credit history.

The goal is to build an engine that correlates transactional behaviour (savings patterns, utility consistency, habitual spendings & etc.,) with demographic stability to produce a single, explainable risk score. The platform must not only score the user but also provide "Explainable AI" (XAI) outputs—showing the why behind the score—and recommend specific financial products tailored to their calculated risk profile. The result should be a bridge between traditional banking requirements and the modern, digital-first lifestyle of loan seekers.

IV. Definition of Done

1. Functional Requirements

- **Multi-Source Data Ingestion** – The solution must ingest and parse three distinct data types: *Transactional Data (CSV/JSON)*, *Demographic Data (JSON)*, and *Product Catalogs (JSON)*.
- **Integrated Scoring Pipeline** – A machine learning pipeline that processes raw inputs, performs feature engineering (e.g., calculating savings ratios or payment consistency, behavioral & habitual info), and outputs a consolidated score.

Rule-Based “New-Age” Credit-Score Framework:

(*Lifestyle + Spending + Repayment Discipline = 0 – 1000 points*)

Score Component	Sub-factor	How to measure	Points range	Example rule
Lifestyle	1.1 Employment Stability	Months at current job (or continuous gig-platform earnings)	0 – 150	If months $\geq 24 \rightarrow +150$; 12 $\leq$ months $< 24 \rightarrow +100$; 6 $\leq$ months $< 12 \rightarrow +50$; else 0
	1.2 Housing Status	Rent-payment history or ownership proof	0 – 80	Own home $\rightarrow +80$; Rent & on-time payments ≥ 12 mo $\rightarrow +60$; Rent < 12 mo $\rightarrow +30$; No stable address $\rightarrow 0$
	1.3 Digital Footprint (mobile-recharge, internet bill regularity)	% of months with on-time payments in last 12 months	0 – 70	$\geq 95\% \rightarrow +70$; 80-94% $\rightarrow +45$; 60-79% $\rightarrow +20$; $< 60\% \rightarrow 0$
	1.4 Education / Skill Level	Highest credential (high-school, bachelor, master, certifications)	0 – 50	Master/PhD $\rightarrow +50$; Bachelor $\rightarrow +40$; Professional cert $\rightarrow +30$; High-school $\rightarrow +20$; $< HS \rightarrow 0$
Spending Behavior	2.1 Spend-to-Income Ratio	(Total monthly spend $\div$ Monthly income)	0 – 120	$\leq 30\% \rightarrow +120$; 30-50% $\rightarrow +80$; 50-70% $\rightarrow +40$; $> 70\% \rightarrow 0$
	2.2 Expense Diversity (mix of essential vs. discretionary)	% of spend on essentials (food, housing, utilities)	0 – 80	Essentials $\geq 70\% \rightarrow +80$; 55-69% $\rightarrow +45$; 40-54% $\rightarrow +20$; $< 40\% \rightarrow 0$

Score Component	Sub-factor	How to measure	Points range	Example rule
	2.3 Cash-flow Volatility (std-dev of monthly net cash flow over 6 mo)	Lower volatility = higher score	0 – 70	$\sigma \leq 5\%$ of income $\rightarrow +70$; 5-10 % $\rightarrow +40$; 10-20 % $\rightarrow +15$; > 20 % $\rightarrow 0$
	2.4 Savings / Emergency Fund	Days of income saved in a liquid account	0 – 80	≥ 180 days $\rightarrow +80$; 90-179 days $\rightarrow +50$; 30-89 days $\rightarrow +20$; < 30 days $\rightarrow 0$
Repayment Discipline	3.1 On-time Payment Rate (all credit, utilities, rent, loans)	% of payments made on or before due date in last 12 months	0 – 200	$\geq 98\%$ $\rightarrow +200$; 95-97 % $\rightarrow +150$; 90-94 % $\rightarrow +100$; 80-89 % $\rightarrow +50$; < 80 % $\rightarrow 0$
	3.2 Debt-to-Income Ratio	Total recurring debt obligations $\div$ Gross monthly income	0 – 120	$\leq 20\%$ $\rightarrow +120$; 21-35 % $\rightarrow +80$; 36-50 % $\rightarrow +40$; > 50 % $\rightarrow 0$
	3.3 Credit Utilization (if any credit lines exist)	Outstanding balance $\div$ Credit limit	0 – 100	$\leq 10\%$ $\rightarrow +100$; 11-30 % $\rightarrow +70$; 31-50 % $\rightarrow +30$; > 50 % $\rightarrow 0$
	3.4 Recent Delinquency Severity	Number of 30-day, 60-day, 90-day+ missed payments in last 24 months.	0 – 150 (negative points)	No delinquency $\rightarrow +150$; 1-30-day miss $\rightarrow +100$; 1-60-day miss $\rightarrow +50$; 1-90-day+ or multiple misses $\rightarrow 0$
Bonus / Penalty Adjustments	4.1 Positive Financial Behaviour (e.g., regular micropayment of a savings app, charitable giving)	Add up to +50	+20 for each qualifying habit (max 3)	
	4.2 Risk Flags (e.g., recent address change, frequent new credit inquiries)	Subtract up to –50	–20 per flag (max –50)	

- **Unified Output –**

For every user processed, the system must emit: **(User Mode)**

- **A Real-time Risk Dashboard:** A visual interface showing the credit score, a breakdown of contributing factors, and risk categorization.

- **Explainability Module:** A text-based or visual "Factor Analysis" (e.g., "Consistency in utility bills +15 pts, Weekday resto -5 pts & etc.,").
- **Automated Recommendation Engine:** A list of financial products (cards, loans) mapped to the user's score.
Here the User can click on the offer for credit card, micro loan and the system connect to the banking system via an API for the user to avail the pre approved loans/ credit cards (Here you are expected to create another application and link both the bank and Credit application via an API key
- **Transparency Report:** A downloadable PDF summary explaining the credit decision (Accept/Reject) for the user.

Business Users Mode: Lender Portal

Enable participating banks, microfinance institutions and fintech lenders to login in as a Business user to browse search and evaluate candidates. The Business user can assess the risk profile and dispatch customer loan and or credit card offers.

- Business User Login with credentials
- Candidate Search and filter engine
 - Credit Score – 650 to
- Anonymization of User personal data – Name, address, phone, until the user accepts the loan offer
- Push offers / campaign – Lenders can select candidates or bulk who qualify and push credit cards or micro loan offers directly to user dashboard

2. Testing & Quality Assurance

- **Data Object Validation** – Implementation of a testing suite to validate score accuracy against a provided synthetic "ground truth" dataset.
- **Integration Test** – A full-stack test that feeds a complete user profile (Demographics + Transactions) and validates the end-to-end flow from API to Dashboard.
- **Robustness Test** – Handling of missing data (e.g., a user with no utility bill history) without system failure.

3. Performance & Accuracy

- **Latency** – The end-to-end scoring process (from data ingestion to dashboard update) must complete in under 15-30 seconds.

4. Code Quality & Reproducibility

- **Modular Architecture** – Clear separation between Data Ingestion, ML Logic, API, and Frontend.
- **Dependency Management** – A single manifest (e.g., **requirements.txt**) to allow judges to run the project instantly.

V. Expected Outcomes

1. **Alternative Risk Engine** – A scoring service that effectively uses non-traditional features (utility consistency, income stability) to predict creditworthiness.
2. **Instant Visual Insight** – An interactive dashboard that:
 - Displays a clear, color-coded credit score (atleast 4 types).
 - Visualizes "Credit Drivers" (Factor Analysis) so the user understands their financial standing.
 - Presents a curated list of "Next Steps" via product recommendations.
3. **Financial Inclusion** – A demonstrable ability to identify "Good" borrowers who would traditionally be rejected due to a lack of credit history.
4. **Explainable AI (XAI) Implementation** – A solution that solves the "Black Box" problem in fintech by providing natural-language explanations for every score.

5. **Scalable Fintech Foundation** – A modular stack (e.g., FastAPI, React, PostgreSQL) that allows for the easy addition of new data types in the future.

VI. Solution Requirements (Good to Have)

1 Core Functional Requirements

- **Advanced Feature Engineering** – Implementation of complex features like "Income Volatility Index" or "Savings Velocity."
- **Dynamic Product Mapping** – A logic engine that automatically updates product recommendations as the user's score changes.
- **Explainability Module** – Using advanced XAI techniques to provide mathematically grounded reasons for score fluctuations.

2 User-Experience & UI Requirements

- **Consumer-Centric Dashboard** – A high-quality, "Fintech-style" UI (like Revolut or Chime) that is intuitive for young users.
- **"What-If" Simulator** – A feature allowing users to see how changes in behavior / factors (e.g., "If I gain steady employment or I take additional debt") would impact their future score.
 - **Saving Buffer Boost**
Input : what if I save extra Rs 1000/- for the next 3 months?
System Impact: Increase the Saving / emergency fund (sub factor 2.4) and reduces cash flow volatility (sub factor 2.3)
Output: Score increases by +35 points, unlocking lower interest rate micro loan
 - **Utility & Rent Consistency Builder**
Input: What if I set up a auto-pay for mu rent and/ or utility bills
System Impact: Maximizes the one-time payment rate (sub factor 3.1) and housing status consistency (sub- factor 1.2)
Output: Score increases by +60 points, qualifying the user for a higher credit limit
 - **Cash flow & Discretionary Sep Over spike**
Input: What if I spend 75% of my monthly income on non essential lifestyle category (dine out, shopping) for 2 consecutive months
System impact: Lowers the expense Diversity ratio (sub factor 2.2) and worsens the spend to income ration (sub factor 2.1)
Output: Score drops by -50 points flagging as account as high cash flow volatility
 - **Utility and Rent Delinquency**
Input: What if I miss my internet bill / mobile / gas/ electricity payment by 45 days and delay the rent by 15 days
System Impact : Triggers Recent Delinquency Severity penalties (sub factor 3.4) and slashes onetime payment rate (sub factor 3.1)
Output: Score drops by -110 points immediately and pre approved credit card offer revoked

Machine Integration

Case 1

In the current scenario, the score is a sum of points (Rule-Based). introduce a "Ground Truth" Dataset of historical users who were either "Successful Borrowers" or "Defaulted."

- **The Complexity Spike:** Instead of calculating a score based on your table, students must train a model (e.g., XGBoost, LightGBM, or Random Forest) to predict the Probability of Default (PD).
Score = $1000 \times (1 - PD)$
- **The Task:** The "Score" (0–1000) should now be a transformation of the predicted probability: .
- **What this tests:** Feature engineering, handling class imbalance (more good borrowers than bad), and model calibration (ensuring a 10% predicted risk actually results in 10% defaults).

Case 2

Propensity to Purchase - Model must build on (or a layer on top of their scoring model) that predicts the probability of conversion for a specific product.

- Input Features:
 - Current Credit Score (from the AltCredit engine).
 - Lifestyle/Spending patterns (e.g., high discretionary spend likely to want a lifestyle card).
 - "What-If" intent (if the user has been playing with the simulator, they are high intent).
- Target Variable: 1 (Will accept offer) or 0 (Will reject/ignore).

Case 3

(Predictive Guidance)

Basis user current score exactly what he/she need to achieve in terms of score so that he / she can get a particular product

- **The ML Implementation:** Instead of just showing the current score, the user sees a "**Target Achievement**" module.
- **The User Experience:**
 - The user selects the "Premium Card" / lending product from the product catalog.
 - The system runs a **Counterfactual Explanation** (an advanced XAI technique). It asks: "*What is the minimum change in my features required to reach the target score?*"
 - **The AI Output:** "*To unlock the Premium Card/lending product, you don't need more income; you need to reduce your 'Cash-flow Volatility' by 12%. Try setting up auto-pay for your electricity bill to improve your 'Repayment Discipline' score.*"
- Must implement logic that doesn't just explain the *past*, but predicts the *future* based on hypothetical feature changes
- **Export Options** – One-click download for the Transparency Report (PDF) and the credit profile (JSON).

3 Non-Functional & Operational Requirements

- **API-First Architecture** – A robust RESTful API that allows the scoring engine to be integrated into external mobile banking apps.
- **Self-Contained Data Store** – Use of an embedded database (SQLite) to ensure the project is "plug-and-play" for judges.

4 Data & Integration Requirements

- **Synthetic-Data Ready** – The solution must accept the exact formats described in the Inputs section without requiring manual pre-processing.

Software Engineering Excellence Checklist

<u>Area</u>	<u>Expectation</u>
<u>Version Control</u>	<u>Code maintained in Git with meaningful commits.</u>
<u>Dependencies</u>	<u>Reproducible build using pinned dependency versions.</u>
<u>Configuration & Secrets</u>	<u>No hard-coded credentials or API keys.</u>
<u>Code Quality</u>	<u>Readable, modular, maintainable code.</u>
<u>Error Handling</u>	<u>Graceful handling of invalid inputs and failures.</u>
<u>Logging</u>	<u>Basic operational logging for troubleshooting.</u>
<u>Testing</u>	<u>Demonstrate positive and negative test scenarios.</u>
<u>Documentation</u>	<u>README explaining setup, execution, architecture, and assumptions.</u>
<u>Build & Run</u>	<u>Solution can be installed and executed using documented steps.</u>
<u>Final Validation</u>	<u>Team demonstrates a clean working build during evaluation.</u>

VII. Detailed breakdown of Tasks

Basic Tasks – Must Have

- **Data Parsers:** Build modules to read and clean the provided Transactional and Demographic datasets.
 - **Core ML Model:** A working Scikit-learn model (Regression or Decision Tree) that produces a numerical score to explain the score.
 - **Basic API:** A FastAPI/Flask endpoint that accepts a user_id and returns a score.
 - **Minimal Frontend:** A web page displaying the Score, the Reason, and the Recommended Product.
- Intermediate

Intermediate Tasks – Must Have

- **Factor Analysis Engine:** Logic that identifies which features (e.g., savings_rate) had the highest impact on the score.
- **Automated PDF Report:** A "Transparency Report" generator that summarizes the credit decision.
- **Database Integration:** Using PostgreSQL or SQLite to store user profiles and historical scores.
- **Simulated "What-If" Analysis: A UI feature where a user can see:** "If I save \$500 more next month, how much will my score increase?"

Advanced Tasks – Good to Have

- **New User Login Portal:** Provision for New user creation. Here a new user can create a profile (Profile creation page). The page should capture User Demographic information. The page should also have upload options where the user can upload Transaction details (pdf/csv)

- **OCR Integration:** Ability to scan a PDF bank statement (using Tesseract) to extract transaction lines for the scoring engine.
- **Deployment:** Containerizing the entire application using Docker.

VIII. Input

1. Transactional Data (CSV/JSON)

- **Bank Statements:** Transaction ID, Date, Amount, Category (Rent, Utility, Salary, Food, etc.), Type (Debit/Credit).
- **Behavioral Indicators:** Timely payment markers for recurring bills (Rent, Internet, Electricity).

2. Demographic Data (JSON)

- **User Profile:** user_id, age, employment_status, education_level, monthly_income, city_tier.
- **Stability Markers:** Length of employment or residency.
- **Lifestyle Data:** Key data that contains all new age details to build the algorithm

3. Product Catalog (JSON)

- **Financial Products:** product_id, name, type (Starter Card, Microloan), min_score_required, interest_rate.

IX. Guardrails

- **No Real PII (Personally Identifiable Information)** – The solution must never ingest, store, or process actual PII, real names, real bank account numbers, or live transaction logs. All inputs provided for this hackathon are strictly synthetic and can be regenerated.
- **No Deep-Learning "Black Boxes"** – Building massive, uninterpretable neural networks is discouraged. The focus is on **Explainability**. Simple, interpretable models (e.g., Decision Trees, Logistic Regression) are preferred so that the "Factor Analysis" remains accurate.
- **No External Credit Bureau APIs** – Do not attempt to call real credit APIs (Experian, TransUnion). Use only the synthetic datasets provided.
- **No Production-Grade Deployment** – The prototype is for demonstration only; it does not require high-availability clustering or complex DevOps.
- **No Legal/Compliance Certification** – The system is a prototype for credit *assessment*, not a legally binding credit decision. It is not expected to meet fully regulatory banking standards.

X. Dataset

Provided along with the Use case Document

XI. Reference

[Alternative Credit Scoring: What is it & How it Works - SEON](#)

[Alternative Credit Scoring: Benefits and Features](#)

[Alternative Credit Scoring in India: Expanding Financial Access | RBL Bank](#)

[Indian banks turn to alternate data sources than credit score for loans to first time borrowers | B...](#)